

# AP Statistics MCQ Practice Exam 02

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## AP Statistics MCQ Practice Exam 2

### PART A, QUESTIONS

#### APSTAT-MCQ2-001

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

On a wildlife reserve, a trail camera records a fox on 28% of nights, a coyote on 19% of nights, and at least one of the two animals on 39% of nights. What is the probability that the camera records both a fox and a coyote on a randomly selected night?

- A) 0.03
- B) 0.05
- C) 0.08
- D) 0.47

#### APSTAT-MCQ2-002

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A high school has a numbered roster of 1,840 students. The principal wants a simple random sample of 60 students for a survey about locker access. Which procedure produces a simple random sample?

- A) Use a random-number generator to select 60 distinct roster numbers from 1 through 1,840.
- B) Randomly select 3 homerooms and survey the first 20 students entering each room.
- C) Survey every 30th student on the roster, beginning with student number 30.
- D) Invite all students to respond and use the first 60 completed surveys.

#### APSTAT-MCQ2-003

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Interpret Results

Prompt:

An acoustics engineer tests  $H_0: \mu = 1.20$  seconds against  $H_a: \mu < 1.20$  seconds, where  $\mu$  is the population mean echo time in rooms fitted with a new wall panel. The test gives a p-value of 0.018. Which interpretation of the p-value is correct?

- A) There is a 1.8% probability that the population mean echo time equals 1.20 seconds.
- B) About 1.8% of fitted rooms have echo times below the sample mean.
- C) If the new panel truly produces a mean echo time below 1.20 seconds, the probability of the observed result is 0.018.
- D) If the population mean is 1.20 seconds, the probability of obtaining a sample mean at least as low as the observed one is 0.018.

APSTAT-MCQ2-004

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

A scatterplot of elevation gain and water consumed for 22 day hikes has a positive linear pattern with  $r = 0.62$ . One hike has much greater elevation gain than all the others. When that hike is removed,  $r$  becomes 0.81. Which statement is best supported?

- A) The unusual hike strengthened the positive linear association and was not influential.
- B) The unusual hike weakened the measured linear association and was influential with respect to  $r$ .
- C) Removing the hike proves that greater elevation gain causes greater water consumption.
- D) The unusual hike must have had a residual of zero because its  $x$ -value was extreme.

APSTAT-MCQ2-005

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

A technician plans a one-proportion  $z$ -test of  $H_0: p = 0.10$  using a random sample of 80 network routers from a population of 5,000 routers. Which condition for the normal approximation is not satisfied under  $H_0$ ?

- A) The sample was selected at random.
- B) The expected number of routers with the characteristic is at least 10.
- C) The expected number of routers without the characteristic is at least 10.

D) The sample size is less than 10% of the population size.

#### APSTAT-MCQ2-006

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

The table shows platform dwell times for 18 trains.

|                      |   |   |   |   |   |
|----------------------|---|---|---|---|---|
| Dwell time (minutes) | 2 | 3 | 4 | 5 | 8 |
| Number of trains     | 2 | 5 | 6 | 4 | 1 |

What is the mean dwell time for these trains?

- A) 3.50 minutes
- B) 4.00 minutes
- C) 4.40 minutes
- D) 3.94 minutes

#### APSTAT-MCQ2-007

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Formulate Questions

Prompt:

An online course awards a distinction to each participant independently with probability 0.35. A researcher wants to simulate the probability that at most 2 of 12 participants earn a distinction. Which simulation design is appropriate?

- A) Let two-digit numbers 00–34 represent a distinction and 35–99 represent no distinction; generate 12 independent numbers per trial and record whether at most 2 represent distinctions.
- B) Let digits 0–2 represent a distinction and 3–9 represent no distinction; generate 12 digits per trial and record whether exactly 2 represent distinctions.
- C) Generate one number from 00–99 per trial and count the trial as a success if the number is 00–34.
- D) Shuffle 12 cards, 2 labeled distinction and 10 labeled no distinction, and draw all cards per trial.

#### APSTAT-MCQ2-008

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

An environmental agency tests  $H_0: p = 0.06$  against  $H_a: p > 0.06$ , where  $p$  is the proportion of private wells in a county with contaminant levels above a safety limit. Which situation describes a Type II error?

- A) The agency concludes that  $p > 0.06$  when the true value is 0.06.
- B) The agency concludes that  $p < 0.06$  when the true value is 0.06.
- C) The agency fails to conclude that  $p > 0.06$  when the true value is 0.09.
- D) The agency fails to conclude that  $p > 0.06$  when the true value is 0.06.

APSTAT-MCQ2-009

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

In a randomized experiment, 150 adults with seasonal allergies receive either a probiotic capsule or an identical-looking placebo. Neither the participants nor the staff members who rate symptom severity know which treatment each participant receives. What is the main purpose of this double-blind procedure?

- A) To reduce placebo effects and observer bias that could systematically affect symptom ratings
- B) To guarantee that the treatment groups have exactly the same symptom severity before treatment
- C) To make the participants a representative random sample of all adults with allergies
- D) To remove natural person-to-person variation from the response variable

APSTAT-MCQ2-010

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

A random sample of 16 adults completed a visual puzzle in a mean time of  $\bar{x} = 72.4$  seconds with  $s = 8.0$  seconds. The sample distribution is approximately symmetric with no outliers. For a 95% confidence interval with  $df = 15$ , use  $t^* = 2.131$ . Which interval estimates the population mean completion time?

- A) (70.3, 74.5) seconds
- B) (64.4, 80.4) seconds
- C) (67.2, 77.6) seconds
- D) (68.1, 76.7) seconds

APSTAT-MCQ2-011

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Formulate Questions

Prompt:

Independent random samples of polymer strips made with formulation A and formulation B are tested for tensile strength. The two population standard deviations are unknown. Which procedure is appropriate for estimating the difference in population mean tensile strength,  $\mu_A - \mu_B$ ?

- A) A one-proportion z-interval
- B) A two-sample t-interval for a difference in means
- C) A paired t-interval for a mean difference
- D) A one-sample t-interval applied to all strips combined

APSTAT-MCQ2-012

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Questions APSTAT-MCQ2-012 through APSTAT-MCQ2-014 use the following stimulus. In a university media archive, 60% of stored files are audio and 40% are video. A stored audio file has probability 0.03 of being corrupted, and a stored video file has probability 0.08 of being corrupted.

What is the probability that a randomly selected stored file is corrupted?

- A) 0.024
- B) 0.032
- C) 0.050
- D) 0.110

APSTAT-MCQ2-013

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Questions APSTAT-MCQ2-012 through APSTAT-MCQ2-014 use the following stimulus. In a university media archive, 60% of stored files are audio and 40% are video. A stored audio file has probability 0.03 of being corrupted, and a stored video file has probability 0.08 of being corrupted.

Given that a randomly selected file is corrupted, what is the probability that it is a video file?

- A) 0.32
- B) 0.40
- C) 0.64
- D) 0.80

#### APSTAT-MCQ2-014

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Interpret Results

Prompt:

Questions APSTAT-MCQ2-012 through APSTAT-MCQ2-014 use the following stimulus. In a university media archive, 60% of stored files are audio and 40% are video. A stored audio file has probability 0.03 of being corrupted, and a stored video file has probability 0.08 of being corrupted.

Are file type and corruption status independent?

- A) No, because the probability of corruption differs between audio and video files.
- B) No, because 60% of files are audio and only 40% are video.
- C) Yes, because the overall probability of corruption is 0.05.
- D) Yes, because every file is either audio or video.

#### APSTAT-MCQ2-015

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

The useful lifetimes of a model of 3D-printer nozzle are approximately normal with mean 860 hours and standard deviation 75 hours. About what lifetime marks the 90th percentile? Use  $z = 1.282$  for the 90th percentile of the standard normal distribution.

- A) 764 hours
- B) 875 hours
- C) 935 hours
- D) 956 hours

#### APSTAT-MCQ2-016

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Collect Data

Prompt:

A county sends a recreation survey to a simple random sample of 600 households, but only 38% of the selected households respond. Why might an interval calculated from the responses fail to estimate the proportion of all county households that use public parks?

- A) The sample size is too large for a one-proportion interval.
- B) Responding households may differ systematically from selected households that did not respond.
- C) Random sampling prevents the use of a confidence interval when the response rate is below 50%.
- D) The interval would describe only the 600 households even if every selected household responded.

#### APSTAT-MCQ2-017

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

Two independent random samples compare the mean number of days required for two compost mixtures to decompose. For mixture A,  $n_A = 20$ ,  $\bar{x}_A = 31$  days, and  $s_A = 4$  days. For mixture B,  $n_B = 18$ ,  $\bar{x}_B = 35$  days, and  $s_B = 5$  days. For  $H_0: \mu_A - \mu_B = 0$ , what is the approximate two-sample  $t$  statistic?

- A) -2.70
- B) -1.35
- C) 0.91
- D) 2.70

#### APSTAT-MCQ2-018

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Interpret Results

Prompt:

Counts of plastic fragments in 25 beach sections form a right-skewed distribution. Twenty-four counts are between 6 and 39, and one count is 94. If the count of 94 is removed, which change is most likely?

- A) The mean increases, while the median and standard deviation both decrease.
- B) The mean and median increase, while the standard deviation decreases.
- C) The mean decreases, while the median and standard deviation both increase.

D) The mean and standard deviation decrease, while the median changes little or not at all.

#### APSTAT-MCQ2-019

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

For wind speeds  $x$  between 5 and 15 meters per second, the line  $\hat{y} = -120 + 48x$  predicts the electrical output  $y$ , in watts, of a small turbine. Which statement about the intercept is most appropriate?

- A) At every wind speed, the turbine loses 120 watts before producing electricity.
- B) A 1-meter-per-second increase in wind speed is associated with a 120-watt decrease in predicted output.
- C) The intercept predicts  $-120$  watts at zero wind speed, but that value is outside the observed  $x$ -range and is not practically meaningful.
- D) The turbine will begin producing electricity precisely when wind speed reaches 120 meters per second.

#### APSTAT-MCQ2-020

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

Researchers test  $H_0: p = 0.12$  against  $H_a: p < 0.12$  for the proportion of patients who experience an adverse reaction. The significance level and the true value of  $p$  under the alternative are held fixed. What is the likely effect of increasing the sample size?

- A) Both the probability of a Type I error and the power decrease.
- B) The power increases because the test can more reliably detect the fixed departure from 0.12.
- C) The power stays the same because  $\alpha$  is unchanged.
- D) The probability of a Type I error increases above  $\alpha$ .

#### APSTAT-MCQ2-021

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

Suppose each of 250 incoming email messages is independently classified as phishing with probability 0.04. Let  $X$  be the number classified as phishing. What are the mean and standard deviation of  $X$ ?

- A)  $\mu_X = 10$  and  $\sigma_X \approx 3.10$
- B)  $\mu_X = 10$  and  $\sigma_X \approx 9.60$
- C)  $\mu_X = 4$  and  $\sigma_X \approx 1.96$
- D)  $\mu_X = 240$  and  $\sigma_X \approx 3.10$

#### APSTAT-MCQ2-022

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

Researchers compare self-reported stress scores of people who commute by electric bicycle with scores of people who commute by car. Participants chose their own commuting method. Even after the groups are similar in age, the e-bike group has a lower mean stress score. Which conclusion is most appropriate?

- A) The study shows an association, but variables such as job location or baseline fitness could confound a causal comparison.
- B) Electric bicycles caused the lower mean because the two groups were compared using the same stress scale.
- C) The lower mean can be generalized to all commuters because participants selected their usual transportation.
- D) A causal conclusion is valid as long as both groups contain at least 30 people.

#### APSTAT-MCQ2-023

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Formulate Questions

Prompt:

A food regulator wants to test whether the population mean mass  $\mu$  of packages labeled 250 grams is less than the labeled amount. Which hypotheses should be used?

- A)  $H_0: \mu < 250$ ;  $H_a: \mu = 250$
- B)  $H_0: \mu = 250$ ;  $H_a: \mu \neq 250$
- C)  $H_0: \mu = 250$ ;  $H_a: \mu > 250$
- D)  $H_0: \mu = 250$ ;  $H_a: \mu < 250$

#### APSTAT-MCQ2-024

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

A 95% confidence interval for  $p_D - p_P$ , the difference between the population proportions of customers who pay digitally and on paper before the due date, is  $(-0.09, 0.02)$ . Which conclusion is supported at the 5% significance level?

- A) Digital payment produces a significantly lower on-time proportion.
- B) Digital payment produces a significantly higher on-time proportion.
- C) The interval does not provide statistically significant evidence of a difference because it includes zero.
- D) The two population proportions are equal because the interval contains zero.

APSTAT-MCQ2-025

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

A botanist measures leaf length in two samples. The 40 oak leaves have mean length 12.5 cm, and the 60 maple leaves have mean length 10.5 cm. What is the mean length of all 100 measured leaves?

- A) 10.9 cm
- B) 11.3 cm
- C) 11.5 cm
- D) 11.7 cm

APSTAT-MCQ2-026

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Interpret Results

Prompt:

An aquarium develops a regression model relating age to body length for a species of fish observed between 2 and 12 months of age. Why should the model not be used to predict the length of a 24-month-old fish?

- A) Correlation can be calculated only when age is measured in years.
- B) A regression line cannot be used to predict a quantitative response.

- C) The prediction is interpolation, so it is guaranteed to be too small.
- D) The prediction is an extrapolation, and the age-length pattern may change beyond the observed range.

#### APSTAT-MCQ2-027

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Collect Data

Prompt:

A playlist contains 30 songs, 8 marked explicit and 22 not marked explicit. Five distinct songs will be played in random order without replacement. Which simulation best estimates the probability that none of the five songs is marked explicit?

- A) Generate five independent random integers from 1 to 30, allowing repeats, and count trials containing only values 9–30.
- B) Generate one random integer from 1 to 30 and count the trial as a success if it is between 9 and 30.
- C) Randomly shuffle 8 E cards and 22 N cards, examine the first five cards, and count trials containing five N cards.
- D) Flip a fair coin five times and count trials with no heads.

#### APSTAT-MCQ2-028

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Formulate Questions

Prompt:

A university independently samples applicants who submitted an optional portfolio and applicants who did not. Let  $p_P$  and  $p_N$  be the respective population proportions who enroll after admission. The university wants to determine whether the enrollment proportions differ. Which hypotheses are appropriate?

- A)  $H_0: p_P - p_N = 0$ ;  $H_a: p_P - p_N \neq 0$
- B)  $H_0: \hat{p}_P - \hat{p}_N = 0$ ;  $H_a: \hat{p}_P - \hat{p}_N \neq 0$
- C)  $H_0: p_P - p_N \neq 0$ ;  $H_a: p_P - p_N = 0$
- D)  $H_0: p_P = p_N$ ;  $H_a: p_P > p_N$

#### APSTAT-MCQ2-029

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

The weekly numbers of pages printed by six members of each of two office teams are shown.

Team A: 18, 19, 20, 20, 21, 22

Team B: 14, 18, 20, 20, 22, 26

Which comparison is correct?

- A) Team A has the larger mean and the larger standard deviation.
- B) The teams have the same mean, but Team B has the larger standard deviation.
- C) The teams have the same mean and the same standard deviation.
- D) Team B has the larger mean, but Team A has the larger standard deviation.

APSTAT-MCQ2-030

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Interpret Results

Prompt:

Independent random samples of two soup recipes are used to estimate  $\mu_N - \mu_C$ , where  $\mu_N$  is the population mean sodium content for the new recipe and  $\mu_C$  is the population mean for the current recipe. A 99% confidence interval, in milligrams per serving, is (1.2, 4.8). Which interpretation is appropriate?

- A) Ninety-nine percent of servings of the new recipe contain 1.2 to 4.8 milligrams more sodium than servings of the current recipe.
- B) The population mean sodium content of the new recipe is estimated to be 1.2 to 4.8 milligrams higher than that of the current recipe.
- C) There is a 99% probability that the sample mean difference lies between 1.2 and 4.8 milligrams.
- D) The recipes have practically important different sodium contents because the entire interval is positive.

APSTAT-MCQ2-031

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

For two events A and B,  $P(A) = 0.40$ ,  $P(B) = 0.25$ , and  $P(A \cap B) = 0.10$ . Which conclusion is correct?

- A) A and B are mutually exclusive because  $P(A \cap B)$  is small.
- B) A and B are dependent because  $P(A) + P(B) \neq 1$ .
- C) A and B are dependent because  $P(A | B) = 0.40$ .
- D) A and B are independent because  $P(A \cap B) = P(A)P(B)$ .

#### APSTAT-MCQ2-032

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

An electric utility tests  $H_0: p = 0.55$  using a random sample of 160 customers, where  $p$  is the population proportion enrolled in automatic payment. The sample proportion is  $\hat{p} = 0.60$ . What is the approximate one-proportion  $z$  statistic?

- A) 1.27
- B) 1.60
- C) 2.02
- D) 2.53

#### APSTAT-MCQ2-033

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

Biologists estimate the mean mass of songbirds in a forest by setting mist nets next to a feeding station at dawn and weighing captured birds. Which concern most directly limits generalization to all songbirds in the forest?

- A) The mass of a captured bird is quantitative rather than categorical.
- B) Measuring at dawn makes random assignment impossible.
- C) Birds that visit the feeding station and enter the nets may differ from birds that do not.
- D) A census is required whenever wildlife can move between locations.

#### APSTAT-MCQ2-034

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

Questions APSTAT-MCQ2-034 through APSTAT-MCQ2-036 use the following stimulus. In a wind-tunnel study of 15 wing models,  $x$  is wing angle in degrees and  $y$  is lift coefficient. Computer output gives  $\hat{y} = 0.420 + 0.065x$  and  $r = 0.867$ .

Which statement correctly interprets the slope of the regression line?

- A) Each additional degree of wing angle causes lift coefficient to increase by exactly 0.420.
- B) For each increase of 0.065 degree, predicted lift coefficient increases by 1.
- C) At a wing angle of 0 degrees, the observed lift coefficient must equal 0.065.
- D) For each additional degree of wing angle, predicted lift coefficient increases by 0.065, on average.

#### APSTAT-MCQ2-035

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

Questions APSTAT-MCQ2-034 through APSTAT-MCQ2-036 use the following stimulus. In a wind-tunnel study of 15 wing models,  $x$  is wing angle in degrees and  $y$  is lift coefficient. Computer output gives  $\hat{y} = 0.420 + 0.065x$  and  $r = 0.867$ .

Another laboratory reports lift index  $L = 100y$  instead of lift coefficient  $y$ . Which equation predicts lift index from wing angle?

- A)  $L = 0.00420 + 0.00065x$
- B)  $L = 0.420 + 6.5x$
- C)  $L = 42.0 + 6.5x$
- D)  $L = 42.0 + 0.065x$

#### APSTAT-MCQ2-036

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Interpret Results

Prompt:

Questions APSTAT-MCQ2-034 through APSTAT-MCQ2-036 use the following stimulus. In a wind-tunnel study of 15 wing models,  $x$  is wing angle in degrees and  $y$  is lift coefficient. Computer output gives  $\hat{y} = 0.420 + 0.065x$  and  $r = 0.867$ .

If wing angle is converted from degrees to radians, with all converted values obtained by multiplying by the same positive constant, what happens to the correlation?

- A) It remains 0.867 because correlation is unchanged by a positive linear rescaling of either variable.
- B) It becomes smaller because numerical angles measured in radians are smaller.
- C) It becomes 0.867 times the degrees-to-radians conversion factor.
- D) It changes to  $-0.867$  because the scale of the explanatory variable changes.

APSTAT-MCQ2-037

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Collect Data

Prompt:

A materials laboratory compares mean breaking strength for tiles fired in kiln A and kiln B. Identical unfired tiles are randomly assigned to a kiln, then their breaking strengths are measured. Why is random assignment important?

- A) It makes the tested tiles a simple random sample of all tiles ever manufactured.
- B) It tends to balance other tile characteristics between kiln groups, supporting a causal comparison of mean breaking strength for the tiles in the experiment.
- C) It guarantees that the two sample standard deviations will be equal.
- D) It allows the laboratory to ignore the independence of individual tile measurements.

APSTAT-MCQ2-038

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

In a very large randomized study, an app notification increased the completion proportion from 71.4% to 72.0%, with  $p$ -value 0.004. Which statement is most appropriate?

- A) The result is not statistically significant because the observed increase is less than 1 percentage point.
- B) The app has a large practical effect because the  $p$ -value is very small.
- C) The null hypothesis has probability 0.004 of being true.
- D) The result is statistically significant, but whether a 0.6-percentage-point increase is practically important requires separate judgment.

APSTAT-MCQ2-039

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Formulate Questions

Prompt:

A nutrition researcher wants to compare distributions of sodium content across breakfast, lunch, and dinner menu categories. Which variable definitions are appropriate for this question?

- A) Sodium content in milligrams as a quantitative variable and meal category as a categorical variable
- B) Sodium content in milligrams as a categorical variable and meal category as a quantitative variable
- C) Both sodium content and meal category as quantitative variables
- D) Both sodium content and meal category as categorical variables

APSTAT-MCQ2-040

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Interpret Results

Prompt:

Each independently processed customer-support ticket has probability 0.12 of being escalated. None of the first 20 tickets today were escalated. What does the law of large numbers suggest?

- A) The next ticket must be escalated so that the proportion begins moving toward 0.12.
- B) Exactly 12 of the next 100 tickets will be escalated.
- C) Over a very large number of tickets, the overall escalated proportion is likely to approach 0.12, but short runs can vary substantially.
- D) The long-run escalation probability has fallen below 0.12 because the first 20 were not escalated.

APSTAT-MCQ2-041

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

Two 95% confidence intervals are constructed for a population proportion. Both samples have the same value of  $\hat{p}$ , but one sample has  $n = 100$  and the other has  $n = 400$ . Assuming all conditions are met, how do their approximate margins of error compare?

- A) The margin for  $n = 400$  is one-fourth the margin for  $n = 100$ .
- B) The margin for  $n = 400$  is one-half the margin for  $n = 100$ .
- C) The margins are equal because  $\hat{p}$  and the confidence level are equal.
- D) The margin for  $n = 400$  is twice the margin for  $n = 100$ .

## APSTAT-MCQ2-042

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

A factory records the defect classification of every item produced during one Tuesday. Which statement best describes the resulting data?

- A) They are a simple random sample of all items the factory will ever produce.
- B) They permit a causal conclusion about why Tuesday defects occurred.
- C) They are a voluntary-response sample because workers chose which items to produce.
- D) They are a census of Tuesday's production, but generalizing to production on other days requires additional justification.

### PART B, ANSWER KEY AND EXPLANATIONS

APSTAT-MCQ2-001, Correct: C, Correct Answer: 0.08, Explanation:  $P(\text{fox} \cup \text{coyote}) = P(\text{fox}) + P(\text{coyote}) - P(\text{both})$ . Therefore  $P(\text{both}) = 0.28 + 0.19 - 0.39 = 0.08$ .

APSTAT-MCQ2-002, Correct: A, Correct Answer: Use a random-number generator to select 60 distinct roster numbers from 1 through 1,840., Explanation: Selecting 60 distinct roster numbers at random gives every possible group of 60 students an equal chance of selection. The alternatives introduce clustering, a fixed systematic start, or voluntary response.

APSTAT-MCQ2-003, Correct: D, Correct Answer: If the population mean is 1.20 seconds, the probability of obtaining a sample mean at least as low as the observed one is 0.018., Explanation: A p-value is computed under  $H_0$  and measures the probability of a test statistic at least as extreme in the direction of  $H_a$ . It is not the probability that  $H_0$  is true.

APSTAT-MCQ2-004, Correct: B, Correct Answer: The unusual hike weakened the measured linear association and was influential with respect to  $r$ ., Explanation: Removing the point raises  $r$  from 0.62 to 0.81, so the point had substantially weakened the measured linear association. The large change shows influence on  $r$ , but it does not establish causation or determine the residual.

APSTAT-MCQ2-005, Correct: B, Correct Answer: The expected number of routers with the characteristic is at least 10., Explanation: Under  $H_0$ , the expected counts are  $np_0 = 80(0.10) = 8$  and  $n(1 - p_0) = 72$ . Because  $8 < 10$ , the large-count condition for routers with the characteristic is not satisfied.

APSTAT-MCQ2-006, Correct: D, Correct Answer: 3.94 minutes, Explanation: The weighted total is  $2(2) + 3(5) + 4(6) + 5(4) + 8(1) = 71$  minutes. Dividing by 18 trains gives  $\bar{x} = 71/18 \approx 3.94$  minutes.

APSTAT-MCQ2-007, Correct: A, Correct Answer: Let two-digit numbers 00–34 represent a distinction and 35–99 represent no distinction; generate 12 independent numbers per trial and record whether at most 2 represent distinctions., Explanation: Thirty-five of 100 equally likely labels model probability 0.35,

12 generated values model the 12 independent participants, and counting trials with 0, 1, or 2 distinctions models “at most 2.”

APSTAT-MCQ2-008, Correct: C, Correct Answer: The agency fails to conclude that  $p > 0.06$  when the true value is 0.09., Explanation: A Type II error is failing to reject  $H_0$  when the alternative is true. Here the agency misses a genuinely elevated well-contamination proportion.

APSTAT-MCQ2-009, Correct: A, Correct Answer: To reduce placebo effects and observer bias that could systematically affect symptom ratings, Explanation: Blinding participants limits response changes caused by knowing the treatment, and blinding evaluators limits differential symptom ratings. It does not create representative sampling or eliminate natural variation.

APSTAT-MCQ2-010, Correct: D, Correct Answer: (68.1, 76.7) seconds, Explanation:  $SE = s/\sqrt{n} = 8/4 = 2$  seconds, so the margin of error is  $2.131(2) = 4.262$  seconds. The interval is  $72.4 \pm 4.262 = (68.138, 76.662)$ , or approximately (68.1, 76.7) seconds.

APSTAT-MCQ2-011, Correct: B, Correct Answer: A two-sample t-interval for a difference in means, Explanation: The response is quantitative, the samples are independent, and the two unknown population means are being compared. Pairing is not present, and a proportion procedure is inappropriate.

APSTAT-MCQ2-012, Correct: C, Correct Answer: 0.050, Explanation: Apply the total-probability rule:  $P(\text{corrupted}) = 0.60(0.03) + 0.40(0.08) = 0.018 + 0.032 = 0.050$ .

APSTAT-MCQ2-013, Correct: C, Correct Answer: 0.64, Explanation:  $P(\text{video and corrupted}) = 0.40(0.08) = 0.032$ . Therefore  $P(\text{video} | \text{corrupted}) = 0.032/0.050 = 0.64$ .

APSTAT-MCQ2-014, Correct: A, Correct Answer: No, because the probability of corruption differs between audio and video files., Explanation: Independence would require  $P(\text{corrupted} | \text{audio}) = P(\text{corrupted} | \text{video}) = P(\text{corrupted})$ . Instead, 0.03 and 0.08 differ, so file type and corruption status are associated.

APSTAT-MCQ2-015, Correct: D, Correct Answer: 956 hours, Explanation: Convert the 90th-percentile z-score back to hours:  $x = \mu + z\sigma = 860 + 1.282(75) = 956.15$  hours, approximately 956 hours.

APSTAT-MCQ2-016, Correct: B, Correct Answer: Responding households may differ systematically from selected households that did not respond., Explanation: A low response rate can create nonresponse bias if park use is related to whether a household responds. The original random selection does not correct systematic differences between respondents and nonrespondents.

APSTAT-MCQ2-017, Correct: A, Correct Answer: -2.70, Explanation: The statistic is  $t = [(31 - 35) - 0] \div \sqrt{[(4^2/20) + (5^2/18)]} = -4 \div \sqrt{2.1889} \approx -4/1.479 = -2.70$ .

APSTAT-MCQ2-018, Correct: D, Correct Answer: The mean and standard deviation decrease, while the median changes little or not at all., Explanation: Removing the extreme high value lowers the mean and greatly reduces spread. Because the median depends on middle positions, deleting one extreme value generally changes it little or not at all.

APSTAT-MCQ2-019, Correct: C, Correct Answer: The intercept predicts –120 watts at zero wind speed, but that value is outside the observed x-range and is not practically meaningful., Explanation: The intercept is the predicted response at  $x = 0$ . Since the observed wind speeds begin at 5 meters per second and negative output is not meaningful here, the numerical intercept should not receive a practical interpretation.

APSTAT-MCQ2-020, Correct: B, Correct Answer: The power increases because the test can more reliably detect the fixed departure from 0.12., Explanation: A larger sample reduces sampling variability, making a fixed alternative value easier to distinguish from the null. With  $\alpha$  fixed, the Type I error rate remains  $\alpha$ .

APSTAT-MCQ2-021, Correct: A, Correct Answer:  $\mu_X = 10$  and  $\sigma_X \approx 3.10$ , Explanation: For  $X \sim \text{Binomial}(250, 0.04)$ ,  $\mu_X = np = 250(0.04) = 10$  and  $\sigma_X = \sqrt{np(1 - p)} = \sqrt{[250(0.04)(0.96)]} = \sqrt{9.6} \approx 3.10$ .

APSTAT-MCQ2-022, Correct: A, Correct Answer: The study shows an association, but variables such as job location or baseline fitness could confound a causal comparison., Explanation: Commuting method was not randomly assigned, so preexisting differences can be related to both method and stress. Matching only on age does not eliminate all plausible confounding.

APSTAT-MCQ2-023, Correct: D, Correct Answer:  $H_0: \mu = 250$ ;  $H_a: \mu < 250$ , Explanation: Equality is placed in  $H_0$  at the labeled amount. The regulator seeks evidence of underfilling, so  $H_a$  is left-tailed.

APSTAT-MCQ2-024, Correct: C, Correct Answer: The interval does not provide statistically significant evidence of a difference because it includes zero., Explanation: Zero is a plausible value for  $p_D - p_P$  within the 95% interval, so the corresponding two-sided test would fail to reject equality at  $\alpha = 0.05$ . This does not prove the proportions are equal.

APSTAT-MCQ2-025, Correct: B, Correct Answer: 11.3 cm, Explanation: The combined mean must weight the group means by their sample sizes:  $[40(12.5) + 60(10.5)]/100 = (500 + 630)/100 = 11.3$  cm.

APSTAT-MCQ2-026, Correct: D, Correct Answer: The prediction is an extrapolation, and the age-length pattern may change beyond the observed range., Explanation: Twenty-four months is well beyond the observed 2-to-12-month range. A linear trend within the data range need not continue for older fish.

APSTAT-MCQ2-027, Correct: C, Correct Answer: Randomly shuffle 8 E cards and 22 N cards, examine the first five cards, and count trials containing five N cards., Explanation: Shuffling the 30 cards and examining five models simple random selection without replacement. Allowing repeats would incorrectly model sampling with replacement.

APSTAT-MCQ2-028, Correct: A, Correct Answer:  $H_0: p_P - p_N = 0$ ;  $H_a: p_P - p_N \neq 0$ , Explanation: The hypotheses concern population proportions, not sample statistics. “Differ” calls for a two-sided alternative, with equality in  $H_0$ .

APSTAT-MCQ2-029, Correct: B, Correct Answer: The teams have the same mean, but Team B has the larger standard deviation., Explanation: Each team is symmetric around 20, so both means equal 20 pages. Team B's values lie farther from 20, so Team B has the larger standard deviation.

APSTAT-MCQ2-030, Correct: B, Correct Answer: The population mean sodium content of the new recipe is estimated to be 1.2 to 4.8 milligrams higher than that of the current recipe., Explanation: The interval estimates  $\mu_N - \mu_C$ , and both endpoints are positive. It concerns a difference in population means, not individual servings, and statistical evidence alone does not establish practical importance.

APSTAT-MCQ2-031, Correct: D, Correct Answer: A and B are independent because  $P(A \cap B) = P(A)P(B)$ ., Explanation:  $P(A)P(B) = 0.40(0.25) = 0.10$ , which equals  $P(A \cap B)$ . This equality is a criterion for independence; events with a positive intersection are not mutually exclusive.

APSTAT-MCQ2-032, Correct: A, Correct Answer: 1.27, Explanation: Under  $H_0$ ,  $SE = \sqrt{[0.55(0.45)/160]} \approx 0.03933$ . Thus  $z = (0.60 - 0.55)/0.03933 \approx 1.27$ .

APSTAT-MCQ2-033, Correct: C, Correct Answer: Birds that visit the feeding station and enter the nets may differ from birds that do not., Explanation: The capture location and method can underrepresent birds that avoid feeders or nets. The resulting convenience-like sample need not represent all forest songbirds.

APSTAT-MCQ2-034, Correct: D, Correct Answer: For each additional degree of wing angle, predicted lift coefficient increases by 0.065, on average., Explanation: A regression slope gives the change in predicted response for a one-unit increase in  $x$ . The wording “predicted” and “on average” avoids an unsupported deterministic or causal claim.

APSTAT-MCQ2-035, Correct: C, Correct Answer:  $\hat{L} = 42.0 + 6.5x$ , Explanation: Because  $L = 100y$ , every predicted response is multiplied by 100. Multiplying both the intercept and slope by 100 gives  $\hat{L} = 100(0.420 + 0.065x) = 42.0 + 6.5x$ .

APSTAT-MCQ2-036, Correct: A, Correct Answer: It remains 0.867 because correlation is unchanged by a positive linear rescaling of either variable., Explanation: Converting degrees to radians multiplies every  $x$ -value by the same positive constant. Correlation measures standardized linear association, so its magnitude and sign remain unchanged.

APSTAT-MCQ2-037, Correct: B, Correct Answer: It tends to balance other tile characteristics between kiln groups, supporting a causal comparison of mean breaking strength for the tiles in the experiment., Explanation: Random assignment reduces systematic treatment-group differences in other tile characteristics. It supports causal inference for the experimental units but does not create a random sample from a broader population.

APSTAT-MCQ2-038, Correct: D, Correct Answer: The result is statistically significant, but whether a 0.6-percentage-point increase is practically important requires separate judgment., Explanation: The  $p$ -value 0.004 is below common significance levels, so the result is statistically significant. Practical importance depends on the size and consequences of the 0.006 increase, not on the  $p$ -value alone.

APSTAT-MCQ2-039, Correct: A, Correct Answer: Sodium content in milligrams as a quantitative variable and meal category as a categorical variable, Explanation: Sodium content is numerical with meaningful arithmetic, while breakfast, lunch, and dinner are group labels. This setup supports comparing a quantitative distribution across categorical groups.

APSTAT-MCQ2-040, Correct: C, Correct Answer: Over a very large number of tickets, the overall escalated proportion is likely to approach 0.12, but short runs can vary substantially., Explanation: The

law of large numbers describes long-run relative frequency. Independence means the first 20 outcomes neither force the next outcome nor change its escalation probability.

APSTAT-MCQ2-041, Correct: B, Correct Answer: The margin for  $n = 400$  is one-half the margin for  $n = 100$ ., Explanation: With  $\hat{p}$  and confidence level fixed, margin of error is proportional to  $1/\sqrt{n}$ . Increasing  $n$  by a factor of 4 multiplies the margin by  $1/\sqrt{4} = 1/2$ .

APSTAT-MCQ2-042, Correct: D, Correct Answer: They are a census of Tuesday's production, but generalizing to production on other days requires additional justification., Explanation: Every item in the defined Tuesday population was recorded, so the data are a census of that day. Tuesday's conditions may differ from other days, so broader generalization is not automatic.

#### SERIES DO-NOT-REPEAT BANK UPDATE, APSTAT-MCQ2

· APSTAT-MCQ2-001 to 005: Unit 2 wildlife-camera fox/coyote events use inclusion-exclusion to recover an intersection from marginal and union probabilities; Unit 1 school locker-access survey selects an SRS by 60 distinct random roster numbers; Unit 4 room echo-time left-tailed mean test interprets a p-value conditionally under  $H_0$ ; Unit 5 hike elevation gain and water use remove an extreme- $x$  observation and assess its influence on  $r$ ; Unit 3 router characteristic test checks null expected counts and finds  $np_0 = 8$  below 10. Misconceptions include confusing union with intersection, treating systematic or voluntary selection as an SRS, interpreting a p-value as  $P(H_0)$ , inferring causation from correlation, and using observed rather than null counts for test conditions.

· APSTAT-MCQ2-006 to 010: Unit 1 train platform dwell-time frequency table computes a weighted mean; Unit 2 course distinctions use a two-digit repeated simulation for a binomial “at most 2 of 12” event; Unit 3 private-well contamination test identifies a Type II error when a true elevated proportion is missed; Unit 1 double-blind probiotic-versus-placebo allergy experiment targets placebo and evaluator effects; Unit 4 visual-puzzle times construct a one-mean t-interval from  $\bar{x}$ ,  $s$ ,  $n$ , and supplied  $t^*$ . Misconceptions include averaging table labels without frequencies, using an incorrect simulation success rate or endpoint, reversing Type I and II errors, treating blinding as random sampling, and omitting  $t^*$  or  $\sqrt{n}$ .

· APSTAT-MCQ2-011 to 015: Unit 4 independent polymer-formulation tensile strengths select a two-sample t-interval; Unit 2 three-question media-archive tree uses audio/video base rates and conditional corruption rates to calculate total corruption probability, Bayes-style  $P(\text{video} \mid \text{corrupted})$ , and dependence from unequal conditional probabilities; Unit 1 normal 3D-printer nozzle lifetimes convert a supplied 90th-percentile z-value back to hours. Misconceptions include treating independent samples as paired, ignoring category base rates, reversing conditional probability, declaring independence from an overall rate, and subtracting rather than adding a positive  $z$  displacement.

· APSTAT-MCQ2-016 to 020: Unit 3 county park-use SRS with 38% response evaluates nonresponse bias; Unit 4 independent compost-mixture decomposition times calculate a two-sample t statistic; Unit 1 right-skewed beach plastic counts reason about removing an extreme high value's effects on mean, median, and standard deviation; Unit 5 turbine wind-speed model interprets an implausible intercept outside the observed range; Unit 3 adverse-reaction proportion test reasons that larger  $n$  increases power at fixed  $\alpha$  and alternative. Misconceptions include assuming random selection cures nonresponse, using the wrong sign or standard error in a two-mean test, treating median as outlier-sensitive, forcing a practical meaning onto an extrapolated intercept, and equating fixed  $\alpha$  with fixed power.

· APSTAT-MCQ2-021 to 025: Unit 2 phishing classifications calculate binomial mean and standard deviation; Unit 1 e-bike versus car commuter stress observational study identifies self-selection and unmeasured confounding; Unit 4 package mass claim formulates a left-tailed one-mean test; Unit 3 digital-versus-paper on-time payment proportions interpret a two-proportion interval that crosses zero;

Unit 1 oak and maple leaf lengths compute a sample-size-weighted combined mean. Misconceptions include confusing variance and standard deviation, claiming causation from an observational comparison, reversing a directional alternative, treating failure to reject as proof of equality, and taking an unweighted average of group means.

· APSTAT-MCQ2-026 to 030: Unit 5 aquarium fish age-length model rejects prediction at 24 months as extrapolation beyond ages 2–12; Unit 2 shuffled explicit/non-explicit song cards simulate five draws without replacement; Unit 3 optional-portfolio applicant enrollment formulates a two-sided difference-of-proportions test with population parameters; Unit 1 two office-team page-count lists have the same mean but different standard deviations; Unit 4 new-versus-current soup sodium means interpret an entirely positive 99% two-mean interval without asserting practical importance.

Misconceptions include treating extrapolation as safe, simulating without-replacement sampling with independent repeated integers, using  $\hat{p}$  in hypotheses, equating equal centers with equal spread, and applying a mean-difference interval to individuals.

· APSTAT-MCQ2-031 to 035: Unit 2 abstract events establish independence from  $P(A \cap B) = P(A)P(B)$ ; Unit 3 utility autopay data calculate a one-proportion z statistic using the null standard error; Unit 1 songbird masses from nets beside a feeder identify selection-related undercoverage; Unit 5 wind-tunnel stimulus interprets the slope of lift coefficient on wing angle and transforms the fitted equation when lift coefficient is multiplied by 100 to create a lift index. Misconceptions include equating non-disjoint events with dependence, using  $\hat{p}$  instead of  $p_0$  in a test standard error, assuming feeder visitors represent all birds, using causal wording for slope, and rescaling only one regression coefficient.

· APSTAT-MCQ2-036 to 040: Unit 5 wind-tunnel output applies the invariance of correlation when degree measurements are converted to radians by positive linear scaling; Unit 4 identical tiles randomly assigned to two kilns connect assignment with balance and causal comparison of mean breaking strength; Unit 3 very large notification experiment distinguishes statistical significance at  $p = 0.004$  from practical importance of a 0.6-point increase; Unit 1 menu analysis identifies sodium as quantitative and meal category as categorical; Unit 2 support-ticket escalations apply the law of large numbers without gambler's-fallacy reasoning. Misconceptions include scaling  $r$  along with measurement units, confusing random assignment with random sampling, using p-value size as effect size, reversing variable types, and believing short-run outcomes must compensate.

· APSTAT-MCQ2-041 to 042: Unit 3 two one-proportion intervals with equal  $\hat{p}$  and  $n = 100$  versus  $n = 400$  use the inverse-square-root sample-size effect to compare margins of error; Unit 1 defect classifications for every Tuesday item distinguish a census of a narrow population from evidence generalizable to other days. Misconceptions include assuming margin scales as  $1/n$  and treating a one-day census as a random sample of all production.

· Unit counts for APSTAT-MCQ2: Unit 1 = 11; Unit 2 = 9; Unit 3 = 9; Unit 4 = 7; Unit 5 = 6. Skill counts: Formulate Questions = 5; Collect Data = 7; Analyze Data = 21; Interpret Results = 9. Stimulus sets: media-archive file-type and corruption tree, APSTAT-MCQ2-012 to 014; wind-tunnel wing-angle regression output, APSTAT-MCQ2-034 to 036.